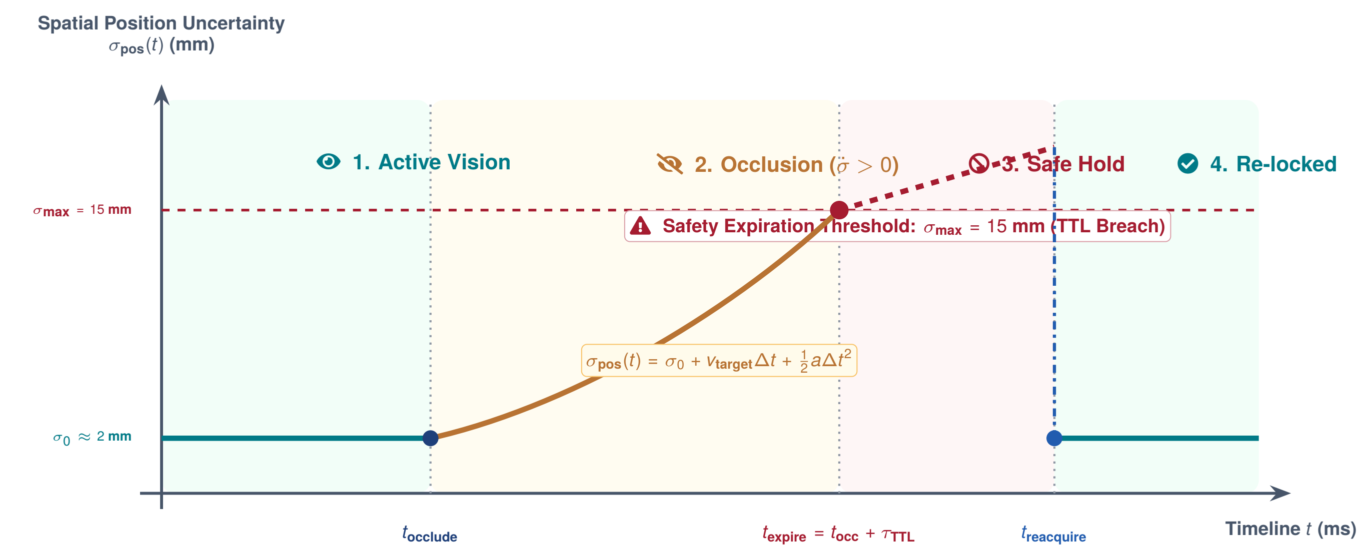
SPATIAL UNCERTAINTY EXPANSION & STATE TTL EXPIRATION

How state covariance inflates during visual occlusion, triggers Time-To-Live (TTL) lease expiry, and recovers upon sensory innovation.



1. NOMINAL TRACKING

- Camera streams 30 Hz tokens
- EKF residual $\|y_t\| \leq 1\sigma$
- Tight bounds: $\sigma_0 \leq 2$ mm
- Full MPU proposal authority

2. OCCLUSION EXPANSION

- Arm occludes line-of-sight
- Covariance $P_t = FP_{t-1}F^T + Q$
- $\sigma(t) = \sigma_0 + v_{\text{tgt}}\Delta t$
- Motion speed derated smoothly

3. TTL LEASE EXPIRY

- $\Delta t_{\text{age}} > \tau_{\text{TTL}}$ (100 ms)
- Spatial bound $\sigma(t) > \sigma_{\max}$
- **MCU Hardware Reflex:** Vetoes proposals; Category 2 velocity hold

4. INNOVATION GATING

- New camera detection arrives
- Mahalanobis check: $d_M \leq 3\sigma$
- **Valid:** Covariance resets $\Sigma \to \Sigma_0$
- $d_M > 3\sigma$: Anomaly rejected

THE CLOSED-LOOP SAFETY CONTRACT

In Physical AI, memory is not an infinite cache. Every state hypothesis carries a decaying confidence lease (τ_{TTL}). When sensory darkness lasts too long, the microcontroller revokes action authority before geometric uncertainty causes physical collisions.